Milestone 1 Submission Report

Formula 1 Podium Prediction | Introduction to Artificial Intelligence

Run provenance: Kaggle notebook ACL_MS1, Version 7 of 7. Kaggle shows a successful 113.1-second run with 110 output files. The run fetched GitHub revision `3f5da97` and used the Formula 1 World Championship dataset (1950-2024). Metrics and figures were extracted from the Version 7 notebook downloaded from Kaggle.

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Kaggle notebook: https://www.kaggle.com/code/crimewizard/acl-ms1
GitHub: https://github.com/CrimeWizard/f1-podium-prediction
Dataset: Formula 1 World Championship, Ergast-derived tables, 1950-2024.

Executive summary

We predict whether a driver finishes on the podium using information available after qualifying and before the race. The notebook uses a temporal train/validation/test split, reports PR-AUC as the primary metric, and includes data audits, three data-engineering analyses, preprocessing comparisons, four model variants, ablation, explainability, and an inference demo. In Version 7, the regularized random forest achieved the highest held-out test PR-AUC (0.696); the shallow neural network achieved the highest validation PR-AUC (0.743). These results are descriptive of this run and split, not guarantees of future performance.

1. Task, data, and prediction cutoff

The prediction unit is one driver-race entry (raceId, driverId). The target is 1 when positionOrder is 1, 2, or 3. Model features are available after qualifying and before the race: actual grid, qualifying position and session times, championship standings entering the race, and prior podium form. Current-race result, status, pit-stop, lap-time, and updated standings fields are excluded from model inputs.

2. EDA, cleaning, and data audit

The 14 CSV tables are loaded as relational inputs. Literal \N sentinels are parsed as missing values. Dates and qualifying lap-time strings receive semantic conversion. Shared-drive duplicates are resolved to one row per (raceId, driverId), retaining the row with the best (lowest) positionOrder. In Version 6, 176 duplicate driver-race rows were found before the full-analysis policy and 131 before the modeling policy. Indianapolis 500 championship rounds (1950-1960) remain in the analysis base for Q1-Q3 and are excluded only from the primary modeling table; a sensitivity experiment compares both policies.

The primary-key audit covered all 14 tables and found no duplicate or null keys. The checked foreign-key relationships had no orphan or null references. The cleaned modeling table contains 26,287 driver-race rows.

EDA before cleaning

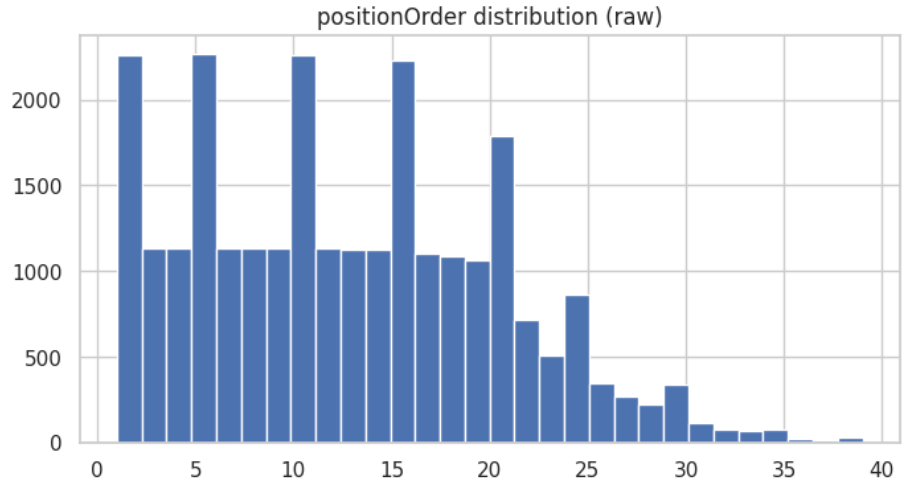


Figure 1. Raw finishing-position distribution.

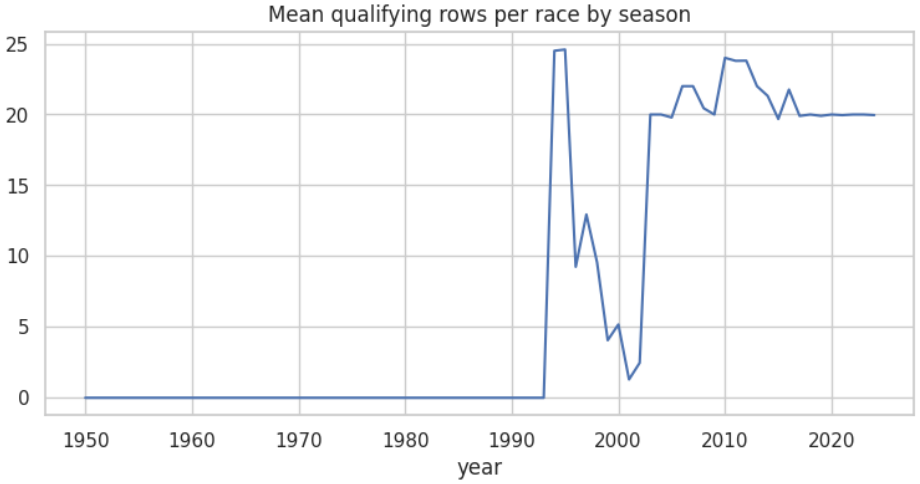


Figure 2. Mean qualifying rows per race by season.

EDA after cleaning

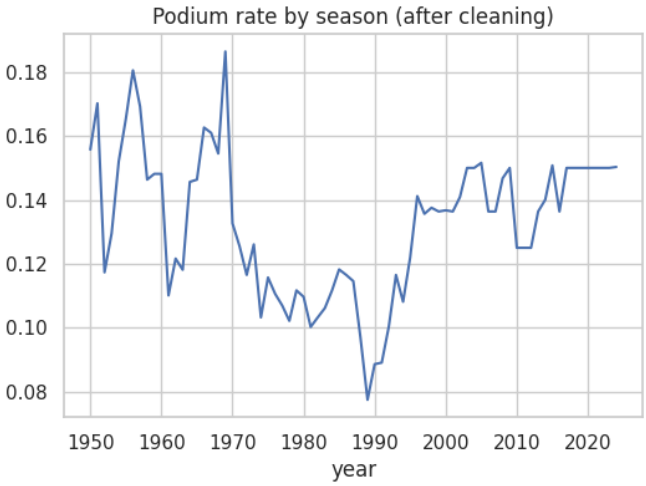


Figure 3. Podium rate by season after cleaning.

3. Data-engineering questions

Q1. Front-row starts and podium conversion

Qualifying positions 1-2 and actual grid positions 1-2 are reported separately because penalties can change the grid. Only starters contribute to denominators; qualifying-based rates also require a valid qualifying position. The notebook excludes 1,601 explicit non-starters and reports 14,612 starters without a valid qualifying position. To reduce small-sample rankings, circuit/era/basis combinations require at least 20 starts. The highest observed rates are Magny-Cours, pre-2014 actual grid (29/36, 80.6%); Yas Marina, from 2014 actual grid (19/22, 86.4%); Suzuka, pre-2014 qualifying (20/26, 76.9%); and Monaco, from 2014 qualifying (17/20, 85.0%). Paired rank correlations across seven shared circuits are -0.180 for actual-grid front rows and -0.291 for qualifying front rows.

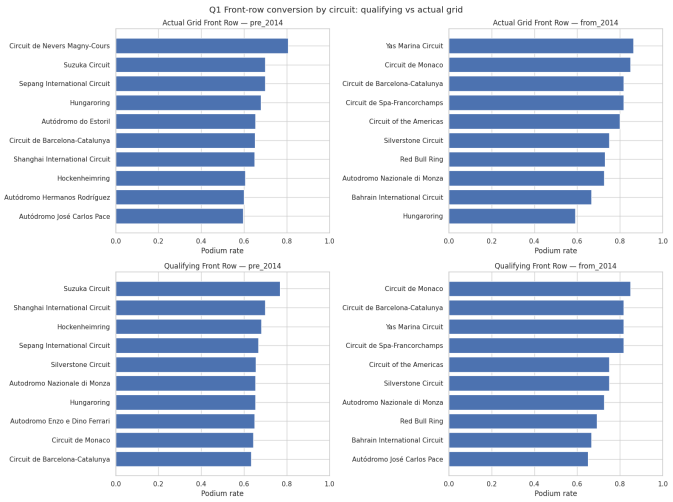


Figure 4. Front-row podium conversion by circuit and era.

Q2. Home-race performance controlling for grid

Home is defined by normalized driver nationality matching the circuit country. The comparison is descriptive within actual-grid bins. Home podium rates are 45.8% (n=430) vs away 45.2% (n=5,180) in grid 1-5; 9.6% (n=479) vs 11.4% (n=5,126) in 6-10; 3.4% (n=469) vs 3.0% (n=5,117) in 11-15; and 0.8% (n=933) vs 0.7% (n=7,203) in 16+. The direction varies by bin, so results do not indicate a consistent home advantage after this control. Unknown-country rows excluded: 0; zero or missing-grid rows excluded: 130.

grid_bin	home_race	starts	podiums	podium_rate
1–5	Away	5180	2340	0.4517
1–5	Home	372	181	0.4866
6–10	Away	5126	582	0.1135
6–10	Home	416	36	0.0865
11–15	Away	5117	154	0.0301
11–15	Home	414	11	0.0266
16+	Away	7202	53	0.0074
16+	Home	729	2	0.0027

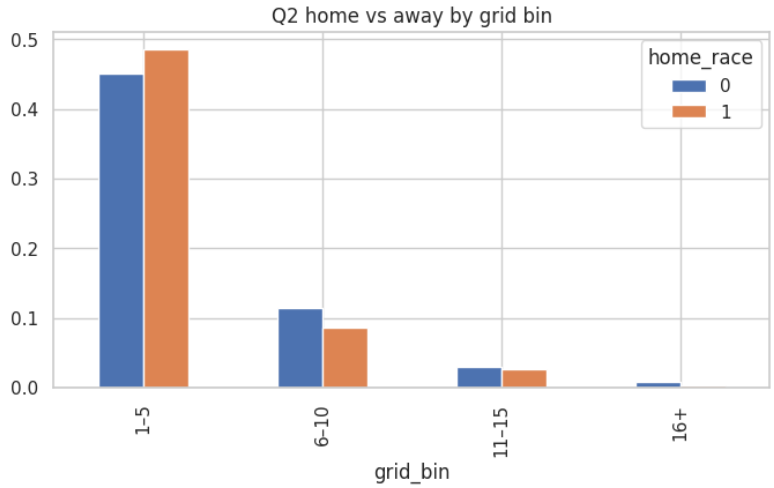


Figure 5. Home versus away podium rates by actual-grid bin (legend: 0 = away, 1 = home).

Q3. Mechanical-retirement rates by constructor

Mechanical retirement categories are based on the documented status mapping. Explicit non-starters are excluded; denominators are starts (including pit-lane starts where the status indicates a start). The highest observed rates are Caterham in 2014-2021 (11/34, 32.4%) and RB F1 Team in 2022-2024 (5/48, 10.4%). These are descriptive rates with modest denominators.

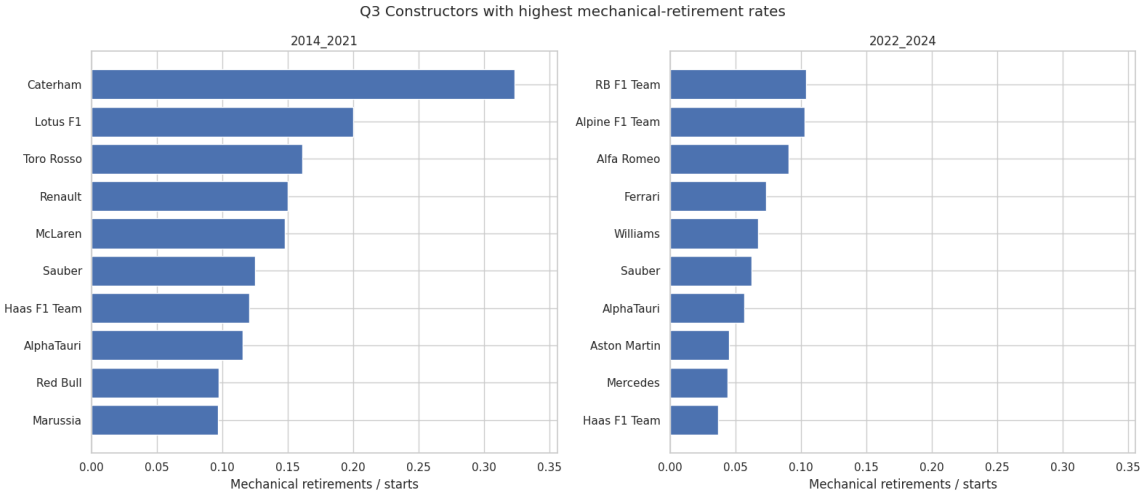


Figure 6. Mechanical retirement rates by constructor and era.

4. Features and leakage control

Feature group	Meaning and pre-race validity
grid	Starting position after penalties; available before the race.
quali_position, q1_sec, q2_sec, q3_sec	Qualifying results available before the race; historical coverage is sparse.
points_before, position_before	Driver championship state shifted to just before this race.
constructor_points_before, constructor_position_before	Constructor championship state before this race.
podium_rate_prev	Rolling podium rate based only on prior races.

The standings features reset at the beginning of each season and are shifted to avoid incorporating the current race. Indianapolis 500 races are retained in full engineering analyses and excluded only from the primary modeling table.

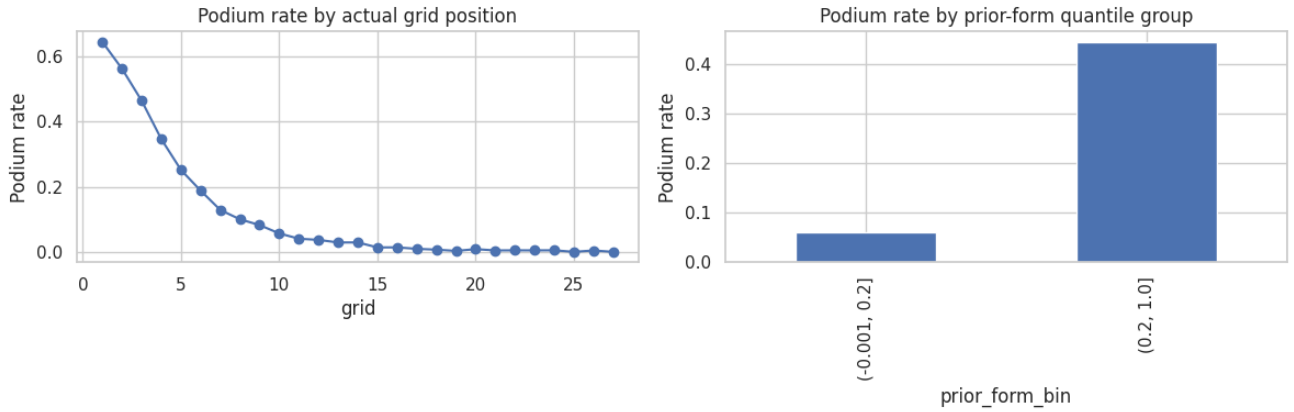


Figure 7. Descriptive relationship of grid and prior form with podium outcomes.

5. Temporal split and preprocessing experiments

The temporal split avoids random shuffling across future seasons: training through 2019 (24,148 rows), validation 2020-2021 (780 rows), and held-out test 2022-2024 (1,359 rows). Podium prevalence was 12.58%, 15.00%, and 15.01%, respectively. Model thresholds are chosen on validation data.

Preprocessing comparison	Validation PR-AUC
Impute then scale	0.7009
Scale then impute	0.7003
Median imputation	0.7009
Mean imputation	0.6906

The two preprocessing orders are nearly tied. Median imputation has a slightly higher validation PR-AUC in this run. Indianapolis policy is included as a modeling sensitivity check.

indy_policy	n_rows	pr_auc_val	pr_auc_test
exclude_indy_500	26287	0.7009	0.6610
include_indy_500	26668	0.7009	0.6620

6. Predictive modeling

PR-AUC is primary because podium outcomes are the minority class. ROC-AUC, F1, accuracy, precision, and recall are also reported for each split. Threshold-dependent metrics use the threshold selected on validation data. Four evaluated variants include logistic regression, default random forest, regularized random forest, and shallow FFNN.

Model	Split	PR-AUC	ROC-AUC	F1	Accuracy	Precision	Recall
Logistic regression	Train	0.5370	0.8914	0.5595	0.8797	0.5189	0.6070
Logistic regression	Validation	0.7009	0.9181	0.6877	0.8987	0.6397	0.7436

Model	Split	PR-AUC	ROC-AUC	F1	Accuracy	Precision	Recall
Logistic regression	Test	0.6610	0.9246	0.6667	0.8837	0.5852	0.7745
Random forest (default)	Train	0.9754	0.9976	0.9535	0.9878	0.9155	0.9947
Random forest (default)	Validation	0.6859	0.9128	0.6900	0.9090	0.7054	0.6752
Random forest (default)	Test	0.6480	0.9250	0.6410	0.8904	0.6303	0.6520
Random forest (regularized)	Train	0.6277	0.9242	0.5945	0.8967	0.5874	0.6017
Random forest (regularized)	Validation	0.7072	0.9237	0.7043	0.9128	0.7168	0.6923
Random forest (regularized)	Test	0.6965	0.9335	0.6420	0.8933	0.6468	0.6373
Shallow FFNN	Train	0.5799	0.9044	0.5725	0.8879	0.5501	0.5968
Shallow FFNN	Validation	0.7433	0.9093	0.7155	0.9154	0.7217	0.7094
Shallow FFNN	Test	0.6624	0.9264	0.6682	0.8926	0.6229	0.7206

The regularized forest has the highest test PR-AUC (0.696). The shallow FFNN has the highest validation PR-AUC (0.743). The default forest has a large train-validation PR-AUC gap, consistent with overfitting; regularization narrows that gap. Validation, not test, is used for model selection and threshold setting.

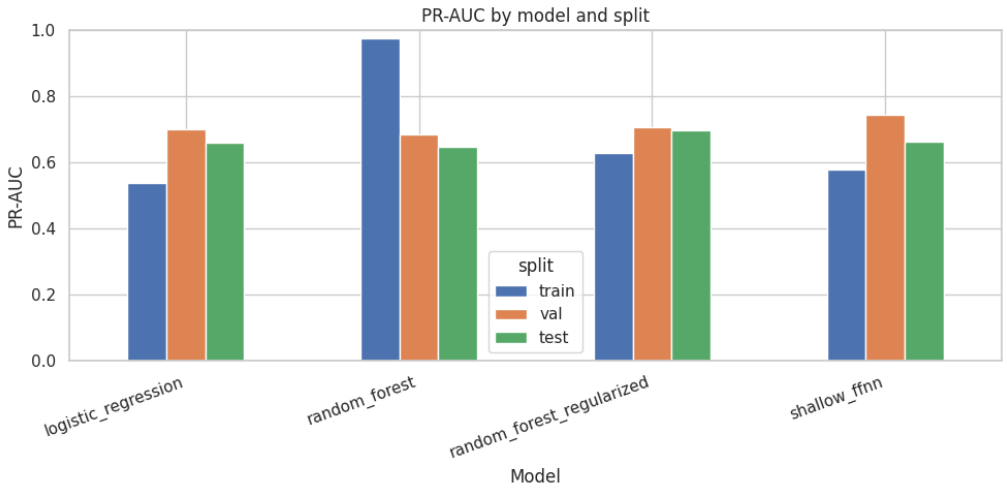


Figure 8. PR-AUC comparison across models and data splits.

7. Feature-group ablation

Each feature group is removed in turn from the logistic-regression baseline; validation and held-out test PR-AUC are compared.

Unnamed: 0	feature_group_removed	pr_auc_val	pr_auc_test
0	none	0.7009	0.6610
1	grid	0.6749	0.6361
2	qualifying	0.6909	0.6509
3	standings	0.7012	0.6565
4	form	0.7071	0.6554

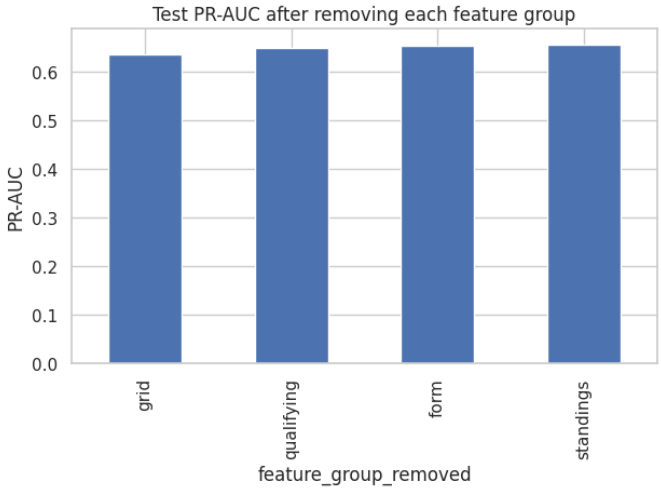


Figure 9. Test PR-AUC after removing each feature group.

8. Explainability

Permutation importance and SHAP provide global descriptions of model reliance; LIME explains one validation prediction. They are not causal evidence. The leading permutation signals are grid and prior podium rate, followed by constructor standing. SHAP used a 100-sample background subset, as noted by the run output.

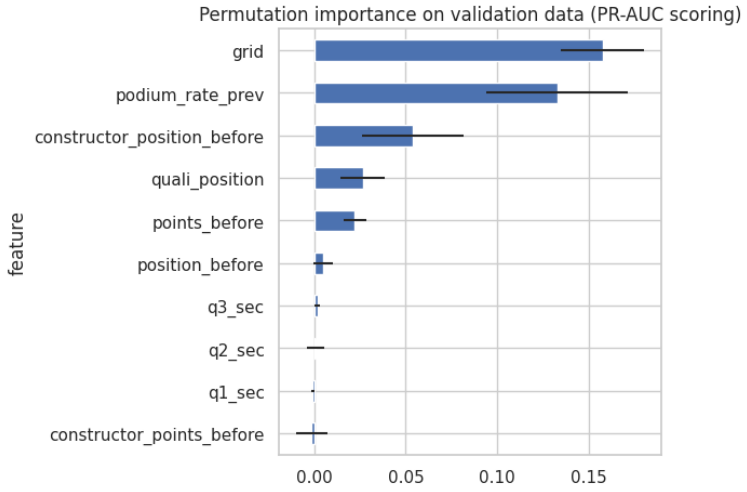


Figure 10. Permutation importance on validation data.

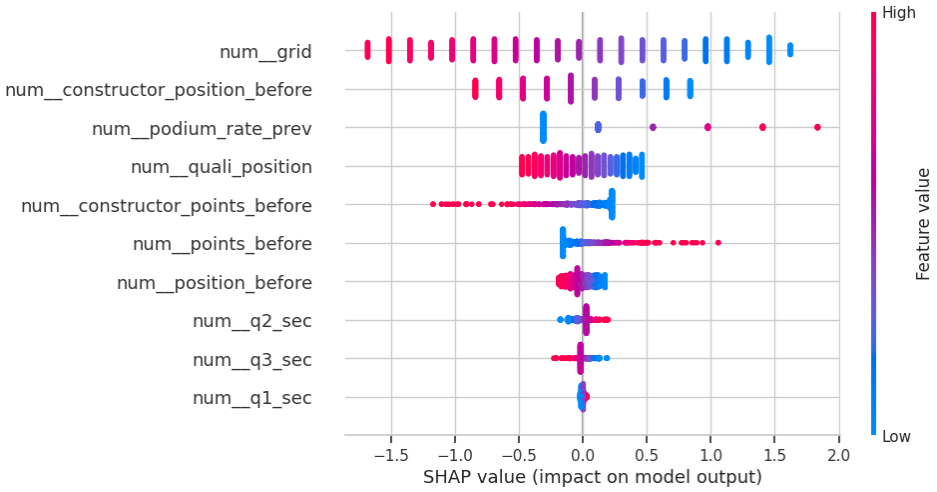


Figure 11. SHAP summary plot.

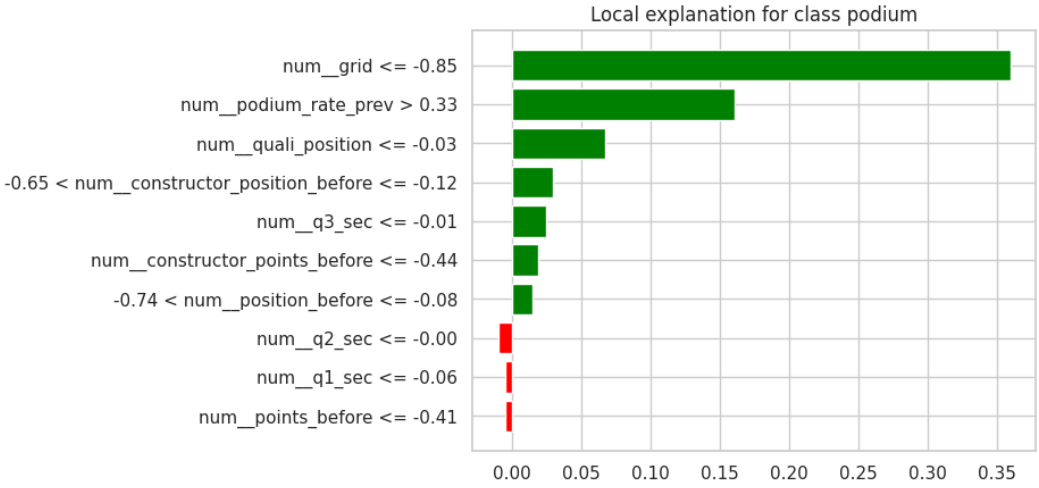


Figure 12. LIME explanation for one validation row.

9. Inference demonstration

The inference helper was demonstrated on a complete validation row and a partial row. The complete row returned podium probability 0.855 and prediction 1; the partial row returned probability 0.405 and prediction 0. The validation-selected logistic threshold was 0.758.

10. Limitations and references

Qualifying coverage is sparse in earlier seasons. Constructor renames complicate historical team continuity. Sprint-specific features are not included. Mechanical-retirement labels depend on status descriptions, and the source has no independent starter flag. Podium outcomes are imbalanced, so PR-AUC is more informative than accuracy alone. Circuit and constructor leaders have limited denominators; all comparisons are descriptive.

References: Milestone 1: Formula 1 Podium Prediction Companion; Formula 1 for Newcomers; Kaggle Formula 1 World Championship dataset; project notebook and GitHub repository linked on page 1.